When Does Deep Reinforcement Learning Help Link Adaptation? A Pre-Registered Evaluation Against Calibrated Rules

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***Abstract*—We study downlink link adaptation (CQI-offset and MCS selection) in a simulated multi-user LTE/NR-like cell with multi-process HARQ, and ask when deep reinforcement learning (RL) beats both outer-loop link adaptation (OLLA) and deterministic rules calibrated on the learner's own cues. The protocol uses pre-registered decision rules and wording, disjoint calibration and evaluation traces, common random numbers, seed-level inference and a public registry of 67 cases. For deadline traffic at an HARQ round-trip time of 8 TTIs, a residual advantage actor-critic (A2C) policy delivers 7.49 % more packets on time than OLLA after a 200-TTI warm-up. Against rules re-calibrated on 500 fresh traces it still wins on evaluation traces, including over a table that combines deadline slack, user class and HARQ-process state (+0.66 %, 90 % CI [+0.29, +1.03] %). That margin is not robust across training seeds (5 of 10 seeds above zero; hierarchical 95 % CI [−0.30, +1.70] %), and re-calibration removes 58 % of the earlier margin over the best rule. The gain is uneven: for users slower than 10 m/s the margin over the v17 Context table is −1.71 %. On full-buffer goodput the learned policy gives no gain (−0.88 % against a calibrated offset). We document three simulation pitfalls that had produced large spurious gains, and release code, traces and the registry.**

***Index Terms*—link adaptation, reinforcement learning, HARQ, evaluation methodology, pre-registration**

I. Introduction

Link adaptation (LA) chooses the modulation and coding scheme (MCS) for each downlink transmission from channel quality indicator (CQI) reports that are delayed and noisy. In LTE and 5G New Radio the industrial baseline is outer-loop link adaptation (OLLA) [1]: an additive offset on the reported CQI that steps down after a NACK and up after an ACK, so that the block-error rate tracks a target. Many recent studies replace or augment OLLA with bandits or deep reinforcement learning (RL) and report clear gains in throughput or reliability over it [2], [3], [4], [5], [6], [7].

A learner usually observes more than OLLA does: deadline slack, delay history, buffer or HARQ state. A gain over OLLA therefore mixes two things, the value of the extra information and the value of *learning* a policy from it. Only a comparator that receives the same information, through a rule calibrated on the learner's cues, separates the two. Of the 23 learning-based LA papers we screened (abstracts and accessible text), 0 compared against such a rule, 3 did so partly, 13 did not, and for 7 the text did not allow a determination. This is not a criticism of those papers, whose settings differ from ours. It leaves open a practical question: *when does deep RL help link adaptation once the comparison includes rules calibrated on the learner's own cues?*

We study this question in one simulator of a multi-user LTE/NR-like cell with multi-process HARQ, for deadline-constrained and full-buffer traffic. The protocol was built up over 17 versions of the campaign and fixed before the final rounds: disjoint calibration, validation and evaluation traces; common random numbers (CRN) [8]; ten training seeds; nine comparators, including OLLA variants and calibrated rules; a fairness guard; headroom bounds; seed-level inference next to trace-level tests; pre-registered decision rules and wording; and a public registry of every case.

The answer is narrower than most of the literature suggests. For deadline traffic with an HARQ round-trip time (RTT) of 8 TTIs, a residual advantage actor-critic (A2C) policy on a calibrated base rule beats OLLA by 7.49 % in on-time delivery after a 200-TTI warm-up. Against rules re-calibrated by exhaustive search on 500 fresh traces it still wins at trace level, but the margin over the strongest one, a table that combines deadline slack, user class and HARQ-process state, is only +0.66 % (90 % CI [+0.29, +1.03] %), and *it is not robust across training seeds*: 5 of 10 seeds are above zero, and the hierarchical seed×trace 95 % interval is [−0.30, +1.70] %. Re-calibration removes 58 % of the margin over the best rule. The gain also depends on the user: for users slower than 10 m/s the margin over the v17 Context table is negative (−1.71 %). A design ablation suggests that the oracle class information in the base rule is not needed, since the same recipe on a non-oracle base beat every comparator; a confirmatory replication of that result on fresh traces is pre-registered. On full-buffer goodput the learner adds nothing: it is −0.88 % against a single calibrated offset, and at RTT 8 it collapses to its base rule.

Our contributions are:

- **C1, protocol and artefacts.** A pre-registered evaluation protocol for learned LA with calibrated comparators, an information table, CRN, disjoint splits, seed-level inference and multiplicity accounting over a registry of 67 cases and 108 scheme evaluations; code, traces and logs are released.
- **C2, a positive case and its limits.** For deadline traffic under realistic HARQ timing, residual A2C beats all calibrated comparators on evaluation traces, including rules re-calibrated by exhaustive search on 500 fresh traces and a table that combines deadline slack, user class and HARQ-process state; the margin over that combined table is not robust to seed resampling (hierarchical 95 %

Draft v1. Items marked as placeholders wait on runs or on later drafting.

CI [−0.30, +1.70] %). The trace-level margin over that table is +0.66 %. The gain is concentrated in faster users; for users below 10 m/s the margin over the v17 Context table is −1.71 %.

- **C3, where RL does not help.** Under full-buffer goodput the learned policy does not beat a calibrated static offset (−0.88 % and −1.73 % vs. Fixed in M3 and M4).
- **C4, pitfalls and recommendations.** Three simulation and design errors produced large apparent gains in earlier versions of this campaign: a mobility time-scale error, slot stealing by a shared reward, and the timing of the proportional-fair (PF) average under multi-process HARQ. We describe them and derive reporting recommendations.

Section II reviews related work. Sections III–V give the system model, the learning method and the protocol. Section VI reports the results and the pre-registered robustness checks, Section VII the pitfalls, and Section VIII the discussion and limitations. The appendices give hyper-parameters, the full comparator tables and the calibrated tables.

II. Related Work

A. Classical link adaptation and OLLA

Outer-loop link adaptation (OLLA) corrects the reported channel quality indicator (CQI) with an offset that steps down after a NACK and up after an ACK, so that the long-run block-error rate approaches a target [1], [9]. Variants adapt the target [10], the step sizes [11], [12], [13], the offset estimator [14], [15], or the feedback used under URLLC soft-ACK [16]. Ahsen et al. use a rank-dependent multi-offset rule [17]; this is a different design from the Multi-OLLA comparator used here. SALAD varies three OLLA step sizes without learning [18].

B. Learned link adaptation

Bandit formulations choose the modulation and coding scheme (MCS) or the offset directly [19], [2], [20], [21]. Deep reinforcement learning (RL) has been applied with DQN, actor-critic, TD3 and PPO agents [3], [22], [4], [23], [7]. Deadline-aware RL on an OLLA loop is studied by Kela et al. [6], and learning on top of OLLA by Pulliyakode and Kalyanasundaram [24], Pang et al. [22] and Wang et al. (LOLLA) [5]. These works are evaluated in different settings from ours (LOLLA, for example, treats uplink with an immediate CRC outcome and no HARQ feedback delay), and we do not re-implement them. Our question is a different one: given a simulator with multi-process HARQ, how much of a learner's gain over OLLA remains once the comparison includes rules calibrated on the learner's own cues?

Table I summarises eight closely related papers on two points that matter for that question. Across the 23 learning-based papers we screened, 0 compared against a calibrated rule built on the learner's cues, 3 did so partly, 13 did not, and 7 could not be determined. These counts rest on abstracts and the text we could access, and an "unclear" entry is not a negative finding.

TABLE I
Eight closely related learned link-adaptation papers. "Calibrated rule": a non-learning rule calibrated on the learner's cues is a baseline. "Parity": the learner and that rule use comparable information. Entries come from the abstract and accessible text.

Paper	Setting	HARQ model	Calibr. rule	Parity
Saxena [2]	single link, AWGN and fading	first transmission only	no	partial
Ye [3]	LTE/NR simulation	not stated	no	unclear
Kela [6]	5G UL, XR, system-level	system-level HARQ	no	no
Pang [22]	LTE/NR simulation	not stated	no	unclear
You [4]	simulation and USRP/srsRAN testbed	parallel HARQ, ACK delay	partial	no
Wang (LOLLA) [5]	GPU NR stack, UL	immediate CRC, no HARQ delay	no	no
Demirel [7]	system-level, multi-cell	system-level	partial	unclear
Wiesmayr (SALAD) [18]	OAI, single user	real HARQ	partial	yes

C. Evaluation rigour

Doherty et al. show that carefully tuned heuristics matched published RL results on optical-network problems [25]; this is the closest methodological precedent, in another domain. General concerns about reproducibility and baselines in learned systems and in wireless machine learning are raised in [26], [27], [28], [29]. For RL specifically, variance across seeds and the reporting of intervals are discussed in [30], [31], [32]. Pre-registration is advocated in [33], [34], and credibility of simulation studies in [35], [36]. We apply these ideas to one concrete question in link adaptation.

III. System Model

All results come from one MATLAB simulator, evaluated in simulation only. Table II lists the parameters, and each is checked against the code by the number-generation script.

A. Cell and channel

A single macro cell serves $N = 10$ users on a 20 MHz carrier at 2 GHz with 100 resource blocks (RBs), 15 kHz subcarrier spacing and a 1 ms transmission time interval (TTI). Large-scale fading is log-normal shadowing with $\sigma = 6$ dB and 50 m decorrelation distance; small-scale fading is Jakes. User speeds are drawn from 3–30 m/s, which gives Doppler frequencies of about 20 Hz to 200 Hz. A two-state Markov process adds interference bursts of 9 dB with mean dwell times of 250 TTIs (calm) and 120 TTIs (burst).

B. CQI feedback and link abstraction

Each user reports a 15-level channel quality indicator (CQI) with a delay drawn from 2–8 TTIs and Gaussian report noise of $\sigma = 1$ dB. Block errors follow an SINR-to-BLER abstraction rather than LDPC link-level tables; this is a limitation (Section VIII). The OLLA reference loop adds an offset Δ to

TABLE II
SIMULATION PARAMETERS (CHECKED AGAINST DEFAULTENV_V17.M AND V17_SETUP.M).

Parameter	Value
Users, bandwidth, RBs	10, 20 MHz, 100
Carrier, subcarrier spacing, TTI	2 GHz, 15 kHz, 1 ms
Shadowing	6 dB, 50 m
Speed range (Doppler)	3–30 m/s (20–200 Hz)
Interference burst	9 dB; dwell 250/120 TTI
CQI delay, noise	2–8 TTI, 1 dB
OLLA target, steps, limit	0.1, 0.5/0.056 dB, 10 dB
HARQ	chase, 2.5 dB, ≤ 3 retx; RTT 8/8 proc. or RTT 0/1 proc.
PF averaging	scheduling time, $\alpha = 0.05$
Deadline traffic	12000 bit / 20 TTI, budget 10 TTI
Trace length, burn-in	200, 50 TTI
Evaluation traces	500 per case

the reported CQI. A NACK lowers Δ by $\Delta_\downarrow = 0.5$ dB, and an ACK raises it by $\Delta_\uparrow = \Delta_\downarrow P_{\mathrm{tgt}}/(1 - P_{\mathrm{tgt}}) \approx 0.056$ dB, for a target $P_{\mathrm{tgt}} = 0.1$, with $|\Delta| \leq 10$ dB. The corrected CQI selects the MCS. We call this corrected CQI the *OLLA reference*. All calibrated rules and the learned action act as CQI-index offsets on top of it.

C. HARQ

Retransmissions use chase combining ($2.5\,\mathrm{dB}$ gain per retransmission, at most 3 retransmissions) and reuse the stored MCS and transport block size. The learner's action therefore affects new transmissions only. We study two timing regimes: *realistic* HARQ with an ACK/NACK round-trip time (RTT) of 8 TTIs and 8 processes per user, and *instantaneous* feedback (RTT 0, one process).

D. Scheduler

A proportional-fair (PF) scheduler gives all RBs to one user per TTI. The PF average rate uses an exponential average with $\alpha = 0.05$ and is updated *at scheduling time* with the scheduled rate. Updating it instead when the ACK arrives is a common alternative; under multi-process HARQ it changes conclusions (Section VII), so we state the choice explicitly (Fig. 1b).

E. Traffic and metrics

Deadline traffic: $12\,000\,\mathrm{bit}$ packets arrive every 20 TTIs with a budget of 10 TTIs. The primary metric is the on-time delivery ratio. *Full-buffer traffic:* the primary metric is goodput. Each trace lasts 200 TTIs, of which the first 50 are burn-in, leaving 150 scored TTIs. We also report the Jain fairness index over per-user means and the 5th-percentile user value (edge5). Four cases form the main campaign: M1 (deadline, RTT 8), M2 (deadline, RTT 0), M3 (goodput, RTT 8) and M4 (goodput, RTT 0).

IV. LEARNING FORMULATION AND METHOD

A. Decision problem

At every TTI each scheduled user receives one decision. The effective CQI is $\mathrm{clip}(q_{\mathrm{ref}} + b_i + \delta)$, where q_{ref} is the OLLA reference, b_i is a calibrated base-rule offset and δ is the learned part. In the *offset* formulation $b_i = 0$ and $\delta \in [-4, +2]$. In the *residual* formulation $\delta \in \{-1, 0, +1\}$ and the base rule is the better of Fixed and FixedPU, chosen on calibration traces (it was FixedPU in all four cases). FixedPU uses the true delay and speed class of each user and is therefore a *class oracle* (Section V).

B. Rewards

For deadline traffic the reward is $+1$ for an on-time and -1 for a missed packet, with potential-based shaping $\Phi(s) = -0.5\,r/B$ for r remaining bits of a B-bit packet [37] and a deviation penalty of $0.02|\delta|$. For goodput the reward is an opportunity cost, the fix for the slot-stealing failure in Section VII.

C. Learner and training

A shared multi-user advantage actor-critic (A2C) [38] uses a 57-128-96-3 actor on three stacked frames of 19 features. It has 20 099 parameters and 19 872 multiply-accumulates per user decision. The logit of $\delta = 0$ starts with a bias of 3. We train for 3000 episodes (deadline) or 800 (goodput) with 10 independent seeds. The *primary scheme* is the per-seed mean: each seed's policy is evaluated and the results are averaged. A deterministic ensemble of the seeds is secondary (pre-registered from v14). Hyper-parameters are in Appendix A. Measured on an i5-8350U CPU, a decision takes 2.03 μs in a batched loop (9.01 μs for a single forward pass), and a full TTI for 10 users 830.3 μs, 2.40 times the FixedPU rule.

V. EVALUATION PROTOCOL

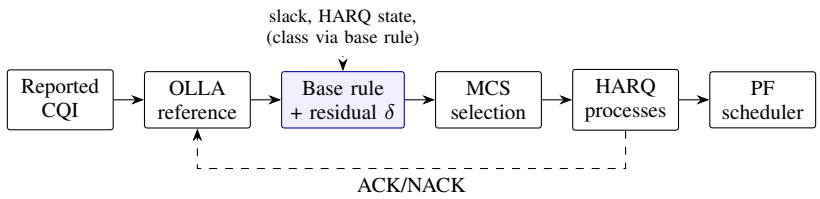
A. Cases and registry

Each round of the campaign uses fresh, disjoint seeds and traces: 500 evaluation traces and 10 training seeds for the headline cases. Every case is recorded in a public registry of 67 cases and 108 scheme evaluations (v9 to v17, Aug–Nov 2026). Of these, 7 cases (18 evaluations) were later superseded after an error was found (Section VII), and 49 cases date from v10, when the present protocol began.

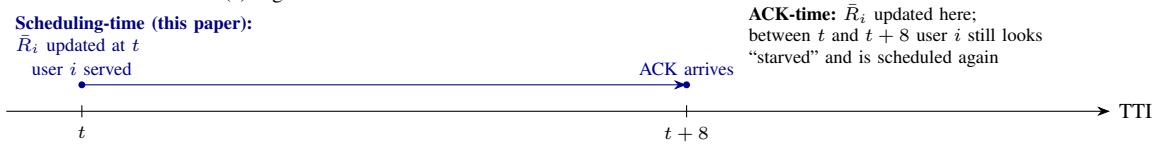
B. Comparators, calibration and information

Nine comparators are evaluated (seven distinct ones in M1, because the tuned-target OLLA equals the default OLLA and the two bandits coincide there). *Baseline*: open-loop CQI. *OLLA*, *OLLA (tuned target)* and *Multi-OLLA* (a fast/slow blend). *Fixed*: one global offset on the OLLA reference. *FixedPU*: one offset per user class on the OLLA reference, where the class is the true delay class (two) and the true mean-speed class (three); this is a *class oracle*, because the true speed is not observable at the gNB. *Context table*: one offset per remaining-slack bin (three bins). Two contextual bandits (global; per delay class) are trained on the learner's episodes.

Table III states what each scheme observes. The learner's state contains only gNB-observable quantities and adds the slack cue and three HARQ-process features. But the residual policy acts on top of FixedPU, so it inherits the class



(a) Signal flow



(b) PF average timing at RTT 8

Fig. 1. (a) Decision path: the learner (shaded) acts as a bounded offset on a calibrated base rule applied to the OLLA reference CQI; the dashed arrow is the OLLA loop. (b) The two PF-average definitions on an RTT-8 timeline.

information of that base rule, including the oracle speed class. For this reason we describe the calibrated rules as *built on the learner's cues* (the slack cue in the Context table, the class information in FixedPU). The G6 stage adds a rule that combines slack, class and HARQ-process state (Section VI-G). With the combined table the comparison becomes a same-information comparison, which the learner wins on evaluation traces but not robustly across training seeds.

C. Calibration of the comparators: three findings

A code review before the final comparison showed that the v17 calibration of the rules was weaker than a "calibrated on the same cues" description suggests. We state three findings plainly.

1) *The v17 Context table was found by a two-pass coordinate ascent started from the all-zero table, on 10 calibration traces.* The search space has $7^3 = 343$ tables (seven offsets in three bins), and the search did not examine all of them.
2) *The v17 FixedPU table was separable.* For each of the nine global offsets one run was made, and each of the six classes took its own best offset from the per-class metrics. The assembled six-entry table was never evaluated as a whole table.
3) *A pre-registered correction (G6) removes both limits.* Fixed (9 candidates), the Context table (343 tables) and the tuned-target OLLA (30 settings) are searched exhaustively on 500 fresh calibration traces (seeds 600001–600500). FixedPU is a separable grid followed by joint refinement (at most three sweeps). A combined table over slack $\times$ delay class $\times$ speed class $\times$ busy HARQ processes (0 or ≥ 1; $3 \cdot 2 \cdot 3 \cdot 2 = 36$ cells) is searched by coordinate ascent from 4 starts. The first start is the *composite* table

$$\mathrm{comp}(b,d,v,u) = \min\big(2, \max(-4, P(d,v) + X(b) - F)\big),$$

with F the re-calibrated Fixed offset, $P(d,v)$ the re-calibrated FixedPU table and $X(b)$ the re-calibrated Context table, all in integer CQI steps. Cells with fewer than 200 calibration decisions are not searched and keep their composite value. The tables are hashed before any evaluation trace is run, and calibration and evaluation values are reported side by side.

The G6 outcomes are reported in Section VI-G.

D. Splits, common random numbers and decision rules

Calibration traces (seeds from 60001), validation traces (from 80001) and evaluation traces (from 500001) are disjoint. All schemes see the same evaluation traces with common random numbers [8], so comparisons are paired per trace. "Scheme A beats comparator X" means a Holm-adjusted [39] two-sided Wilcoxon signed-rank $p < 0.05$ [40] *and* a lower bound > 0 for the 90 % confidence interval of the paired percentage margin. *Claim A*: beats the OLLA family. *Claim B*: beats Fixed, FixedPU and the Context table. A *WIN* means beating all nine comparators. A fairness guard applies: a Jain-index drop above 0.02 or an edge5 drop above 10 % relative to the best calibrated rule must be reported. Two headroom bounds frame the results: a per-trace hindsight bound and a genie bound.

Trace-level intervals treat seeds as fixed. We therefore also report seed-level inference: a hierarchical seed$\times$trace bootstrap [41] (20,000 resamples), a bootstrap over the ten per-seed means, and a one-sided sign test over seeds. Under the pre-registered rule, a claim that passes at trace level but whose seed-level lower bound is not above zero is reported as "not robust to seed resampling".

E. Pre-registered robustness programme

After the v17 results, and before any further run, we froze eight short pre-registrations that test the M1 result from different sides (Table IV). Each text fixes the question, the data, the comparator family, the decision rule and the wording for every outcome. The texts and the analysis scripts are hashed, and the freeze commits are recorded with the code. A change after freezing is allowed only before the affected stage starts, and is logged as a dated amendment with its reason;

TABLE III
INFORMATION USED BY EACH SCHEME. ✓: USED; –: NOT USED; "ORACLE": SIMULATOR GROUND TRUTH, NOT A GNB MEASUREMENT.

Scheme	CQI	ACK hist.	Slack	Delay cls.	Speed cls.	HARQ state	Other	Calibrated on
Baseline (open loop)	✓	–	–	–	–	–	–	–
OLLA	✓	✓	–	–	–	–	–	– (standard)
OLLA, tuned target	✓	✓	–	–	–	–	–	target grid, 10 traces
Multi-OLLA	✓	✓	–	–	–	–	CQI trend, cell NACK average	–
Fixed	✓	✓	–	–	–	–	–	offset grid, 20 traces
FixedPU (class oracle)	✓	✓	–	✓	✓ *oracle*	–	–	per-class grid, 20 traces
Context table	✓	✓	✓(3 bins)	–	–	–	–	coord. ascent, 10 traces
Bandit (global)	✓	✓	–	–	–	–	–	learner's episodes
Bandit (delay class)	✓	✓	–	✓	–	–	–	learner's episodes
A2C, offset action	✓	✓	✓	–	–	✓	ΔCQI, observed SINR, OLLA offset, 3 frames	800 ep. × 10 seeds
Residual A2C (primary)	✓	✓	✓	✓ (base)	✓ *oracle (base)*	✓	as offset A2C	3000 ep. × 10 seeds
Combined table (G6)	✓	✓	✓(3 bins)	✓	✓ *oracle*	✓ (busy)	–	coord. ascent, 500 traces
Residual A2C, Fixed base (NR1)	✓	✓	✓	–	–	✓	as offset A2C	3000 ep. × 10 seeds

two such amendments changed frozen texts, both before any result of the affected stages had been read, and a third added NR2. Every stage runs a short preflight that measures its cost and pauses the pipeline if the cost exceeds the budget by more than half. Calibration tables are written and hashed before any evaluation trace is touched. Each stage writes its own summary file, and the manuscript reads its numbers from those files through generated macros. A correction to a generated verdict is made in the claims ledger and in the text, never in the generated file (Section VI-G gives one example).

F. Pre-registration and multiplicity

From v12 on, a pre-registration note was written before each run, and failed predictions are reported. In v17 the primary predictions were met in all four cases, and two secondary predictions on M1 were not: the 800-episode offset policy was predicted to win and failed Claim B, and the 800-episode residual policy was predicted to fail Claim B and won. Across the registry, 9 of the 60 valid cases show a WIN under the corrected verdict. Multiplicity is handled over the whole registry: the M1 comparison against the best rule has Holm-adjusted $p = 9.1 \times 10^{-13}$, and $p = 8.8 \times 10^{-10}$ after a Bonferroni factor of 972 (108×9 comparators). We state plainly that the recipe was chosen during the campaign; M1 is a confirmatory test of that recipe on fresh data.

VI. RESULTS

A. Deadline traffic with realistic HARQ (M1)

The residual A2C policy (primary scheme) delivers 61.26 % of packets on time in M1, against 60.31 % for the Context table, 60.14 % for Fixed, 59.65 % for FixedPU and 54.72 % for OLLA (Fig. 2, Table V). Its margin is +11.96 % over OLLA, +12.16 % over Multi-OLLA, +2.70 % over FixedPU, +1.87 % over Fixed and +1.58 % over the Context table (90 % CI [+1.23, +1.94] %; Holm $p = 9.1 \times 10^{-13}$). The policy beats all nine comparators under the rule of Section V. In absolute terms the margin over the Context table is +0.95 percentage points. The tuned-target OLLA coincides with the default

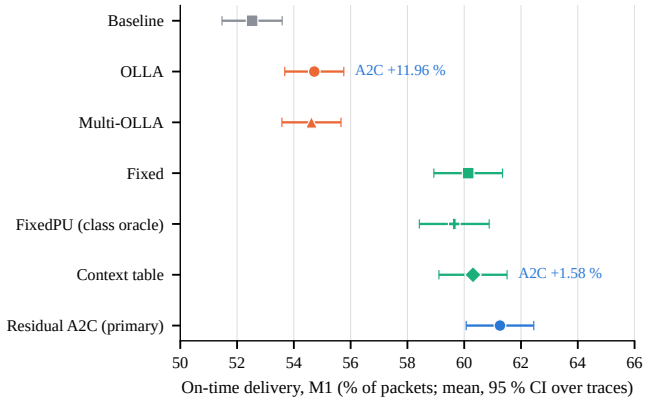


Fig. 2. M1 on-time delivery ratio by scheme (mean and 95 % CI over 500 traces). Blue: residual A2C; orange: OLLA family; green: calibrated rules; gray: baseline. Annotations give the margin of the primary scheme.

OLLA in M1 (the target 0.1 was selected on the calibration traces), and the two bandits coincide.

How much of the gain over OLLA do simple calibrated rules already capture? A single calibrated offset (Fixed) captures 83 % of the learner's gain over OLLA, and the Context table captures 85 %. A comparison against OLLA alone would therefore overstate what the learner adds. The comparators in this subsection are the v17 rules, calibrated on 20 (Fixed, FixedPU) or 10 (Context table, tuned OLLA) traces; Section VI-G replaces them with rules re-calibrated on 500 traces.

B. Seed-level robustness

Trace-level intervals treat the ten training seeds as fixed. Seed-level views are therefore reported next to them (Fig. 4). Against the Context table, 9 of 10 seeds lie above zero; the exception is seed 2 (-0.80 %), and the one-sided sign test gives $p = 0.0107$. The 95 % interval for the mean margin is [+0.72, +2.53] % by bootstrap over the ten seed means, [+0.48, +2.68] % by a t-interval, and [+0.62, +2.62] % by the hierarchical seed×trace bootstrap. Against Fixed the corresponding intervals are [+1.01, +2.81] %, [+0.77, +2.98] %

TABLE IV
PRE-REGISTERED ROBUSTNESS PROGRAMME (FROZEN BEFORE ANY OF THESE STAGES RAN). "FRESH": TRACES OR SEEDS NOT USED IN ANY EARLIER ROUND.

Stage	Question	Data	Decision rule	Status
G1	What does the policy do, and where?	logged M1/M3 decisions, all seeds	descriptive; a mechanism sentence needs a bin $\geq 5\,\%$ of decisions and a pattern in every seed	run
W1	Is the gap to OLLA a cold-start effect?	M1 traces, OLLA warmed up for 200 TTIs	Claim A/B as in M1; the abstract quotes the smaller gain	run
G6	Does the result survive stronger comparators?	500 fresh calibration traces; M1 evaluation traces	Holm over 10 comparators; Claim A, B-500, joint parity; Rule R2 for the best comparator	run
NR1	Is the class oracle in the base rule needed?	fresh seeds, non-oracle Fixed base; M1 evaluation traces	Holm over nine; WIN = beats all	run
G10	Does it hold at other operating points?	two neighbouring cells, fresh traces; single-cue rules re-calibrated per cell	Holm over nine per cell; ALL / MIXED / NONE	running
A1, A3	By user speed; inference cost	M1 traces; timing on one CPU	descriptive	run
A6	Is the multi-process HARQ gain a PF-timing artefact?	100 fresh traces, three schemes	expectation stated in advance	run
NR2	Does NR1 replicate on fresh data?	fresh seeds, fresh evaluation traces, frozen G6 tables	trace level (Holm) and seed level (R2) vs. the combined table	frozen, after G10

TABLE V
M1 COMPARATOR TABLE (V17 COMPARATORS). MARGIN: PAIRED PERCENTAGE GAIN OF RESIDUAL A2C OVER THE COMPARATOR, WITH THE 90 % CI OVER TRACES AND THE HOLM-ADJUSTED WILCOXON p. "BEATS": THE PRE-REGISTERED RULE IS MET. RE-CALIBRATED COMPARATORS ARE IN TABLE VI.

Comparator	On-time (%)	Margin (%)	90 % CI (%)	Holm p	Beats
Baseline	52.53	+16.62	[+15.74, +17.50]	2.8×10^{-76}	yes
OLLA	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
OLLA (tuned target)	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
Multi-OLLA	54.62	+12.16	[+11.40, +12.92]	3.1×10^{-67}	yes
Bandit (global)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Bandit (delay class)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Fixed	60.14	+1.87	[+1.52, +2.23]	1.3×10^{-16}	yes
FixedPU (class oracle)	59.65	+2.70	[+2.36, +3.03]	1.3×10^{-31}	yes
Context table	60.31	+1.58	[+1.23, +1.94]	9.1×10^{-13}	yes

and [+0.89, +2.90] %. Against these v17 rules all seed-level lower bounds are above zero. The hierarchical 90 % interval is about 2.3 times wider than the trace-only one, which is the price of counting seed variance. Section VI-G shows that this changes against the strongest re-calibrated comparator.

C. Fairness and cost

The gain does not come at the expense of fairness under the guard. Against the Context table the Jain index differs by -0.0025 (0.7715 against 0.7740), and edge5 is +1.2 % higher (0.2148 against 0.2122). Goodput is 4.4506 Mbps against 4.3635 Mbps.

D. Replication without HARQ delay (M2)

With instantaneous feedback (M2) the margin over the Context table is +0.43 % (90 % trace interval lower bound +0.21 %), and the hierarchical 95 % interval is [-0.03, +0.87] %, so M2 is described as a replication only: 8 of ten seeds lie above zero. Earlier RTT-0 wins (v13, v14, v16) used the other PF-averaging definition and are not used as evidence. Appendix B gives the full M2–M4 comparator tables.

E. Goodput: no gain (M3, M4)

On full-buffer goodput the primary scheme does not beat calibrated rules: $-0.88\,\%$ (M3) and $-1.73\,\%$ (M4) against Fixed. At RTT 8 the residual policy is bit-identical to its base rule in 5000 of 5000 seed×trace cells: on all 379380 scheduled new transmissions it chose $\delta = 0$ (100.0 %). At RTT 0 this is not the case (2985 of 5000 cells are identical), and the policy does worse than its base rule.

F. Headroom and training length

How much room is there? The per-trace hindsight bound over Fixed is +3.4 % in M1 (+2.0 % in M2), and a genie bound +37.8 % (Fig. 5a). The learner's M1 margin of +0.95 points over the Context table is small against these bounds. Training length matters: at 800 episodes the margin over Fixed is +0.62 % (90 % CI [+0.34, +0.91] %) and at 3000 episodes it is +1.87 % ([+1.52, +2.23] %) (Fig. 5b). The 800-episode variant passes Claim B at trace level, but only 6 of ten seeds lie above the Context table.

G. Robustness checks

Four pre-registered checks ask whether the M1 result survives stronger comparators (G6), a warmed-up OLLA (W1), a

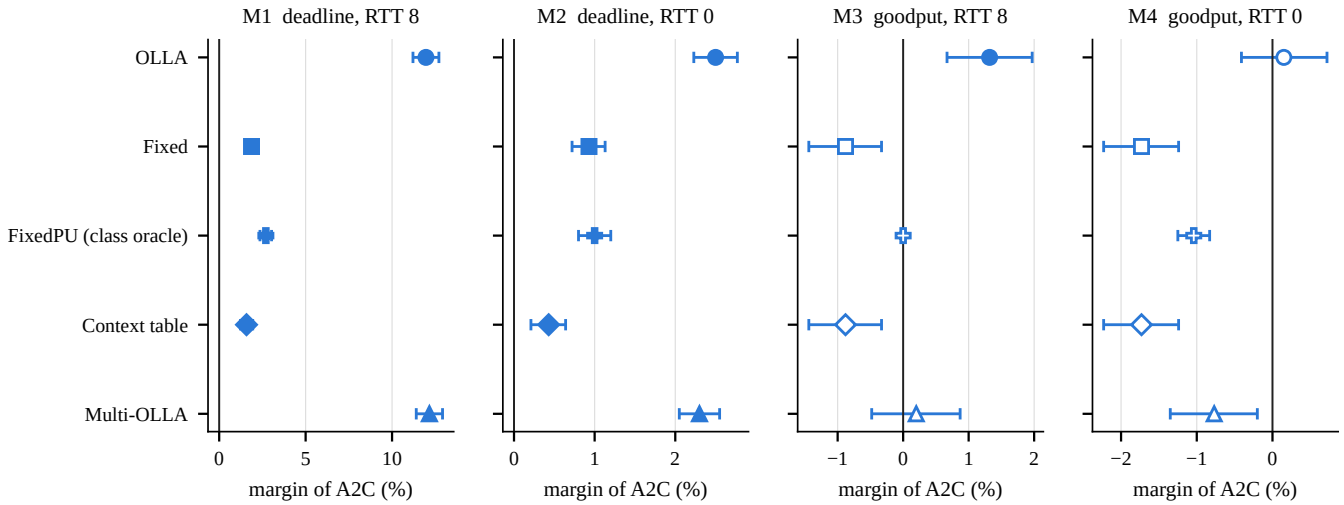


Fig. 3. Margin of residual A2C over each comparator in the four cases (90 % CI over traces). Filled markers: the pre-registered "beats" rule is met; open markers: it is not. The two bandits are omitted for scale (see Table V).

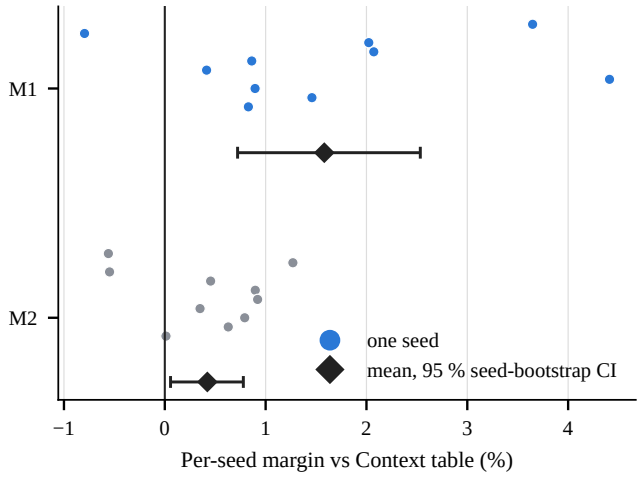


Fig. 4. Per-seed margin over the v17 Context table for M1 and M2 (one dot per training seed) with the mean and its 95 % bootstrap interval over seed means (diamond).

base rule without oracle information (NR1) and other operating points (G10). Their texts were frozen before any of them ran.

1) Comparator calibration budget (G6): In v17 the rules were calibrated on 20 or 10 traces, and two searches were incomplete (Section V). G6 re-calibrates every rule on 500 fresh calibration traces (seeds 600001–600500), disjoint from the evaluation traces. Fixed, the Context table and the tuned-target OLLA are searched exhaustively; FixedPU gets a separable search with joint refinement. G6 adds a *combined table* (Context-joint-500) over deadline slack, delay class, speed class and busy HARQ processes: 36 cells, searched by coordinate ascent from 4 starts, with 21 cells holding at least 200 calibration decisions and the others keeping a composite value (Appendix C). This table uses the same cues as the learner's state and base rule, including the oracle speed class. All tables were hashed before any evaluation trace was run. The combined table scores 0.6088 on the calibration traces and 0.6086 on the evaluation traces; for Context-500 the values are 0.5982 and 0.6040. There is therefore no sign of over-fitting to the calibration set.

Table VI and Fig. 6a give the result. At trace level residual A2C beats all 10 comparators under Holm correction over the family: +1.43 % over Context-500, +1.87 % over Fixed-500, +2.70 % over FixedPU-500 and +8.40 % over the re-tuned OLLA. Over the combined table the margin is +0.66 % (90 % CI [+0.29, +1.03] %, Holm $p = 1.0 \times 10^{-3}$), or +0.40 percentage points (0.6126 against 0.6086). Claim A, Claim B-500 and joint parity therefore all pass at trace level.

Seed level. Against the combined table only 5 of 10 seeds have a positive margin (one-sided sign test $p = 0.623$). The bootstrap over seed means gives [−0.19, +1.60] % and the hierarchical bootstrap [−0.30, +1.70] % (95 %). Both include zero, so Rule R2 applies: *the margin over the combined table is not robust to seed resampling.* Against every single-cue rule the hierarchical lower bound stays above zero (Context-500: [+0.47, +2.46] %). The stage's own summary file marks the claim as seed-robust because it evaluated R2 against Context-500 only; R2 requires the best comparator, and we correct the verdict accordingly (the file itself is left unchanged). The secondary deterministic ensemble of the ten seeds reaches 0.6306 on-time; it was not part of the G6 decision rule and is reported only to show how much of the seed spread an ensemble removes.

How much margin did calibration remove? Against the v17 Context table the margin was +1.58 %; against the combined table it is +0.66 %. Re-calibration and the combined table thus remove 58 % of the margin over the best rule. Most of what looked like a learning advantage over the v17 rules is available to a deterministic table that is calibrated carefully on the same cues.

2) OLLA warm-up (W1): OLLA starts each trace cold, and only 150 TTIs are scored. W1 lets all loops run for 200 TTIs

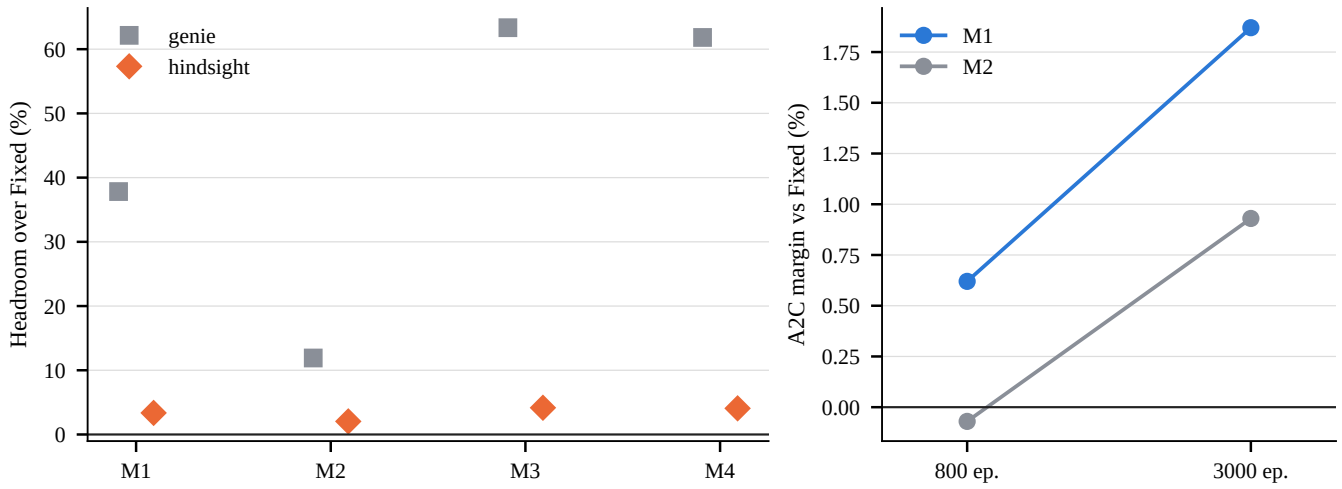


Fig. 5. (a) Headroom over Fixed in the four cases: genie and per-trace hindsight bounds. (b) Margin of residual A2C over Fixed after 800 and 3000 training episodes.

TABLE VI
G6: RESIDUAL A2C (M1 PRIMARY) AGAINST COMPARATORS RE-CALIBRATED ON 500 FRESH TRACES, ON THE 500 M1 EVALUATION TRACES. TRACE LEVEL: MARGIN, 90 % CI AND HOLM-ADJUSTED p OVER THE FAMILY OF 10. SEED LEVEL: SEEDS WITH A POSITIVE MARGIN, HIERARCHICAL SEED×TRACE 95 % CI, AND RULE R2 ("ROBUST": THE LOWER BOUND IS ABOVE ZERO).

	Trace level				Seed level		
Comparator	Margin (%)	90 % CI (%)	Holm p	Beats	Seeds > 0	Hier. 95 % CI (%)	Robust
Baseline	+16.62	[+15.74, +17.50]	3.2×10^{-76}	yes	–	–	–
OLLA	+11.96	[+11.21, +12.71]	1.5×10^{-66}	yes	–	–	–
OLLA-tuned-500	+8.40	[+7.74, +9.07]	6.0×10^{-55}	yes	10/10	[+7.15, +9.66]	yes
Multi-OLLA	+12.16	[+11.40, +12.92]	3.6×10^{-67}	yes	–	–	–
Fixed-500	+1.87	[+1.52, +2.23]	2.0×10^{-16}	yes	9/10	[+0.91, +2.92]	yes
FixedPU-500 (class oracle)	+2.70	[+2.36, +3.03]	1.7×10^{-31}	yes	10/10	[+1.74, +3.76]	yes
Context-500	+1.43	[+1.08, +1.77]	6.1×10^{-11}	yes	9/10	[+0.47, +2.46]	yes
Bandit (global)	+46.55	[+45.22, +47.88]	1.5×10^{-82}	yes	–	–	–
Bandit (delay class)	+46.55	[+45.22, +47.88]	1.5×10^{-82}	yes	–	–	–
Combined table (Context-joint-500)	+0.66	[+0.29, +1.03]	1.0×10^{-3}	yes	5/10	[−0.30, +1.70]	**no**

before scoring. The margin over OLLA falls from +11.96 % to +7.49 % (90 % CI [+6.79, +8.19] %), so part of the cold-start gap was OLLA convergence. The learner itself barely changes (0.6126 cold, 0.6116 warm). Against the calibrated rules the warm-up leaves the result intact: +1.75 % over the Context table ([+1.41, +2.10] %) and +2.08 % over Fixed, so Claim B passes under warm-up. Following the pre-registration, the abstract quotes the smaller (warm) gain over OLLA.

3) Non-oracle base rule (NR1): The M1 policy acts on FixedPU, which uses the true speed class. NR1 retrains the same recipe on the non-oracle Fixed base (one global offset, −2 CQI steps, equal to the Fixed-500 offset), with fresh training seeds and, by design, M1's evaluation traces and comparators. The learner's own state is gNB-observable in both cases, so NR1 uses no oracle information at all. NR1 delivers 0.6235 on time and beats all 9 comparators: +3.38 % over the Context table (90 % CI [+3.02, +3.75] %), +4.52 % over FixedPU and +13.95 % over OLLA (Fig. 6b). It is also +1.77 % above the M1 oracle-base policy ([+1.55, +2.00] %, $p = 2.0 \times 10^{-31}$). The gain does not require class-oracle information in the base rule.

A secondary, descriptive comparison on the same traces puts NR1 +2.44 % above the combined table of G6 ([+2.04, +2.85] %, Wilcoxon $p = 1.5 \times 10^{-19}$). Two cautions apply. NR1 was evaluated on M1's traces, so it is a design ablation and not an independent replication. And the NR1 stage stored only the seed-mean value per trace, so Rule R2 cannot be applied to it. A confirmatory replication with fresh evaluation traces, fresh seeds and per-seed outputs is therefore pre-registered (NR2). NR2 had not been run at the time of writing.

4) Operating points (G10): G10 tests two neighbouring operating points (a higher load at the M1 deadline, and a lower load with a looser deadline) against the OLLA family and the single-cue rules re-calibrated per cell; the combined table is tested at the M1 point only (Amendment 2). The result is established at one operating point (12 kbit packets, 10-TTI budget); other loads and deadlines were not tested under the corrected scheduler.

H. Policy behaviour

1) Action shares (G1): On 587016 scheduled new transmissions in M1 the residual policy chooses $\delta = +1$ in 54.6 %,

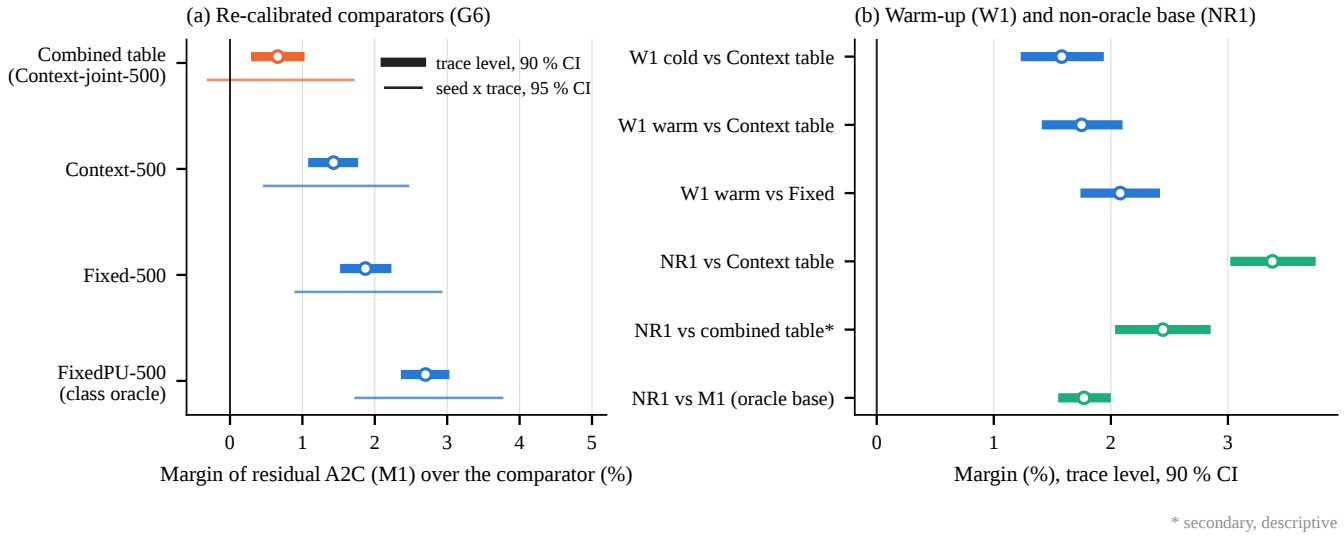


Fig. 6. Robustness checks. (a) G6: margin of residual A2C over the re-calibrated comparators; thick bars: trace-level 90 % CI; thin bars: hierarchical seed×trace 95 % CI. Orange: the seed-level interval includes zero (Rule R2). (b) W1 (OLLA warm-up of 200 TTIs) and NR1 (non-oracle Fixed base); trace-level 90 % CIs. NR1 has no seed-level interval because per-seed values were not stored.

$\delta = 0$ in 26.4 % and $\delta = -1$ in 19.0 % of decisions; clipping cancels some actions, so the effective offset differs from the base rule in 63.4 % (Fig. 7b). The share of $\delta = +1$ varies widely between seeds (11–78 %), so the ten policies are far from identical even though their on-time ratios are close. Three patterns hold in every seed. The $+1$ share is higher with one or two busy HARQ processes than with none (84.7 % against 37.0 %; higher in 10 of 10 seeds; 36 % of decisions), when less than 0.6 of the deadline budget remains (89.6 % against 48.4 %; 10 of 10 seeds; 15 % of decisions), and during interference bursts (69.6 % against 48.7 %; 10 of 10 seeds). These patterns are consistent with a policy that accepts a higher block-error risk for a larger transport block when retransmissions are already pending or the deadline is close. We have not tested this mechanism, for example by attributing on-time packets to state bins. On M3 every decision is $\delta = 0$ (100.0 %), as noted above.

2) Results by user speed (A1): A1 splits the M1 evaluation by each user's mean speed and compares the primary policy with the *v17 Context table* (calibrated on 10 traces); the G6 tables were not part of A1. The margin is +2.82 % for users between 10 and 20 m/s (hierarchical 95 % CI [+1.80, +3.89] %; 47 % of user-traces) and +2.25 % above 20 m/s ([+0.91, +3.68] %; 32 %). For users slower than 10 m/s the margin is *negative*: −1.71 % ([−3.32, −0.15] %; 21 % of user-traces) (Fig. 7a). The policy's actions hardly depend on speed (the $+1$ share is 51.9, 54.3 and 56.7 % from the slowest to the fastest class). This is consistent with a largely speed-independent correction that helps when the channel ages quickly between report and use and over-shoots when it ages slowly. Again, the mechanism was not tested.

VII. Pitfalls and Recommendations

Three simulation or design errors changed conclusions during the campaign. We describe them because each can recur in other link-adaptation studies.

A. Mobility time scale

Early versions (up to v9) moved users v metres per TTI instead of $v\,T_{\mathrm{tti}}$. Channel dynamics were then about a thousand times too fast. Under this error the learner appeared to beat OLLA; after the correction, the v9 registry shows 0 wins in 18 cases, and the A2C policy was −4.08 % against OLLA on goodput.

B. Slot stealing

A per-user goodput reward let a shared agent lower one user's rate to benefit others' slots. The learned policy was bimodal (58 % of actions at -3 and 35.8 % at $+2$), and goodput was 1.38 Mbps against 2.92 for OLLA. An opportunity-cost reward removed the effect (2.9191 against 2.9233 Mbps, a -0.14 % difference that is not significant).

C. PF-average timing under multi-process HARQ

If the PF average is updated with ACKed bits at $t + \mathrm{RTT}$, a just-served user looks starved for RTT TTIs and is scheduled again. In a six-trace diagnostic, Baseline goodput at RTT 8 rose from 3.466 to 4.837 Mbps when going from one to eight processes (+39.6 %) while the Jain index fell from 0.503 to 0.420. Re-running all 13 schemes with the other PF definition changed goodput by 10–26 %, but that comparison is confounded by traces, drain and useful-bit accounting, so it is quoted only as a range. A 10.71 % engine discontinuity had the same cause. The sign of a simple comparison depends on the definition: open-loop Baseline was +4.5 % above OLLA under delivered-throughput PF (v9) but OLLA is +7.4 % above Baseline under scheduling-time PF in M3 (and +6.9 % in M4). A controlled demonstration on 100 fresh traces (A6, Fig. 8)

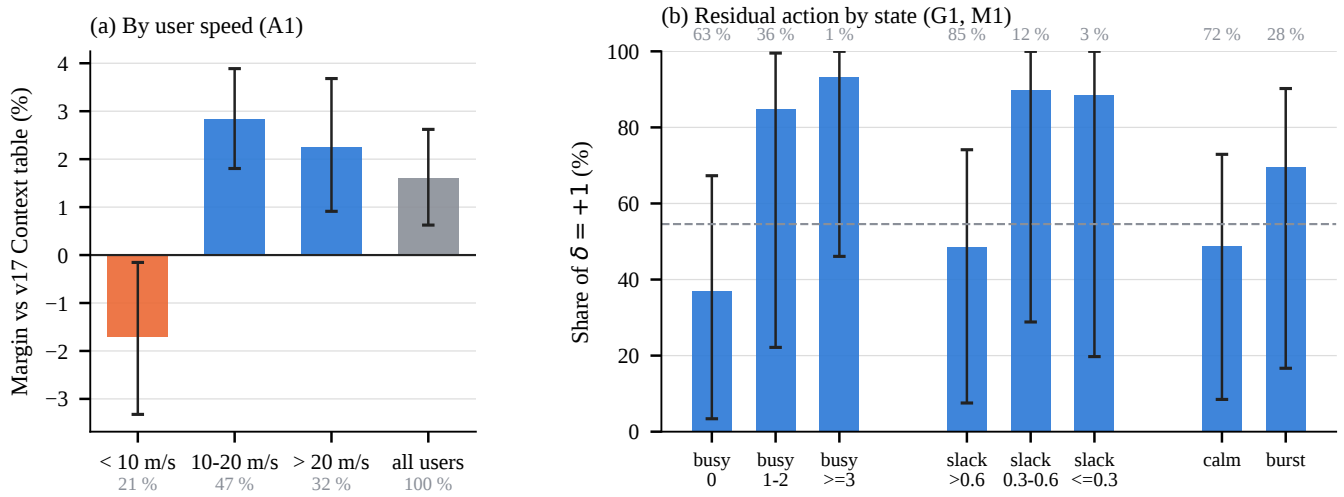


Fig. 7. Policy behaviour in M1. (a) A1: margin of residual A2C over the v17 Context table by user mean speed, with hierarchical 95 % CIs; gray labels give the share of user-traces. (b) G1: share of $\delta = +1$ on scheduled new transmissions by state bin, pooled over seeds (bars) with the minimum and maximum over the ten seeds (whiskers); gray labels give the share of decisions in each bin; dashed line: all decisions. Slack is the remaining fraction of the deadline budget.

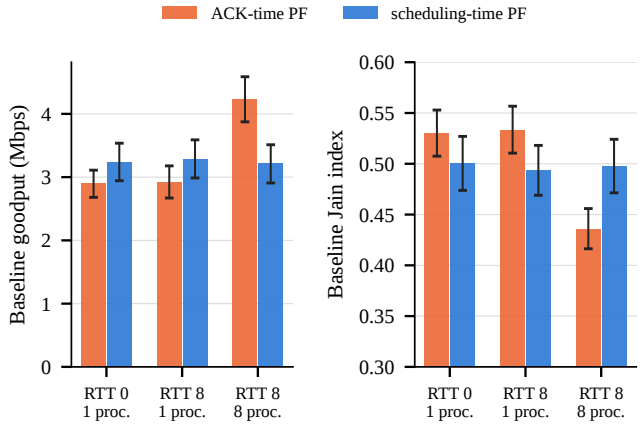


Fig. 8. A6: Baseline goodput and Jain index under the two PF-average definitions, for RTT 0 with one process and RTT 8 with one or eight processes (mean and 95 % CI over 100 traces).

isolates the effect. Under ACK-time averaging, going from one to eight processes at RTT 8 raises Baseline goodput by a factor of 1.447 (2.92 to 4.23 Mbps) and lowers the Jain index from 0.534 to 0.436. Under scheduling-time averaging the factor is 0.976 and the Jain index stays at 0.498 (from 0.494). OLLA and Fixed behave the same way (factors 1.433 and 1.420 under ACK-time averaging, 0.976 and 1.007 under scheduling-time averaging). The pre-registered expectation held for all three schemes: in this diagnostic, the timing of the PF average, not HARQ itself, accounts for the apparent gain.

D. Withdrawn and superseded results

Results from v7 and v8 are withdrawn. A v10 exploratory win (E4c) used overlapping traces. The v12 pre-registered replication failed Claim B and the v13 X1 prediction failed. The v15 and v16 RTT>0 cases (7 cases, 18 evaluations) are superseded because they used the ACK-time PF average. The registry in the supplement lists every case.

E. Recommendations

1) Compare against rules calibrated on the learner's cues, state the calibration budget, and give an information table such as Table III.
2) Use common random numbers, disjoint calibration, validation and evaluation traces, and a public case registry.
3) Report seed-level intervals next to trace-level ones.
4) Report results by mobility regime, HARQ timing and observation set. The observation-set ablation in LOLLA (Table V(a) of [5], 254, 185 and 171 Mbps) shows how much a result can depend on what the learner sees.
5) State the PF-averaging definition and report the complexity of the learner.

VIII. Discussion and Limitations

A. When does RL help link adaptation?

In this study RL helped only where the decision had structure that a static offset cannot use. On full-buffer goodput (M3, M4) every useful input is already summarised by a calibrated offset per user class, and the residual policy learned to leave its base rule alone at RTT 8 and did worse than it at RTT 0. With deadline traffic and delayed HARQ feedback (M1) the policy found something to add. Its corrections are most frequent when HARQ processes are busy, when little of the deadline budget remains and during interference bursts (Section VI-H). This is consistent with the policy trading block-error risk against the chance of finishing a packet in time, a trade-off that a table indexed by a few bins approximates only coarsely. We did not test this mechanism.

Three qualifications bound the positive result. First, the margin over the strongest calibrated rule is small: +0.66 %,

or +0.40 percentage points of on-time delivery. Second, it is not robust across training seeds: only 5 of 10 seeds beat the combined table, and the hierarchical interval $[-0.30, +1.70]$ % includes zero. The ten policies also act very differently (the share of upward corrections ranges from 11 to 78 % across seeds). A practitioner who trains one policy cannot expect to land on the favourable side of the combined table. Third, the gain is not uniform across users. Against the v17 Context table, users above 10 m/s gain (+2.82 % and +2.25 %), while users below it have a margin of -1.71 % (hierarchical 95 % CI $[-3.32, -0.15]$ %), and these slow users make up 21 % of user-traces. The policy's actions barely depend on speed, so a speed-aware residual or a speed-gated deployment might remove the loss; we have not tried either.

B. Calibrate first

The largest effect in this study is not the learner but the comparator. Against OLLA started cold the learner wins by +11.96 %; a warm-up of 200 TTIs reduces this to +7.49 %. One calibrated offset (Fixed) captures 83 % of the learner's gain over OLLA, and the v17 Context table captures 85 %. Exhaustive re-calibration on 500 traces and a combined table remove a further 58 % of the remaining margin. Each step is cheap compared with training ten deep policies, and each changes the conclusion a reader would draw. We therefore read the M1 result as: *calibrate first*; at the studied point the learned residual then adds a fraction of a percentage point on average, with a seed-dependent sign against the best table.

The calibration study also says something about the base rule. On the evaluation traces the non-oracle Fixed-500 offset scores +0.82 % above the class-oracle FixedPU-500 table, although FixedPU was the base rule selected for M1 on 20 calibration traces. A base rule picked on a small calibration set can thus be the weaker one even when it has more information. This is consistent with NR1, where the residual policy on the Fixed base was +1.77 % above the M1 policy on the FixedPU base. NR1 used M1's evaluation traces and has no seed-level analysis, so it does not establish a better configuration. The pre-registered replication NR2 (fresh traces, fresh seeds, the frozen combined table as primary comparator) decides that question.

C. Cost

The actor has 20 099 parameters and 19 872 multiply-accumulates per decision. On an i5-8350U laptop CPU a decision takes 2.03 μs in a batched loop, and a full TTI for 10 users 830.3 μs, 2.40 times the cost of the FixedPU rule in the same simulator. This is small, but training is not: 3000 episodes per seed and ten seeds, against a table search that needs no gradient steps.

D. Limitations

1) *Simulator.* One macro cell, $N = 10$ users and one user scheduled per TTI. There is no multi-cell interference coordination, carrier aggregation or frequency-selective scheduling. Block errors use an SINR-to-BLER abstraction rather than NR LDPC link-level tables, and there is no cross-validation in ns-3 or Sionna. All results come from one MATLAB code base.
2) *Seed robustness.* The margin over the combined table is not robust to seed resampling (Rule R2). NR1 cannot be assessed at seed level because per-seed values were not stored.
3) *Heterogeneous gain.* For users below 10 m/s the margin against the v17 Context table is -1.71 %. The A1 breakdown was not repeated against the G6 tables.
4) *Operating point.* The result is established at one operating point (12 kbit packets, 10-TTI budget); other loads and deadlines were not tested under the corrected scheduler.
5) *Oracle information.* The M1 base rule FixedPU uses the true speed class, which the gNB does not observe, and the combined table uses it too. The learner's own state is gNB-observable. NR1 removes the oracle from the base rule but reuses M1's evaluation traces; NR2 is pending.
6) *Scoring window.* Traces score 150 TTIs after a burn-in. OLLA starts cold unless warmed up (W1).
7) *Selection over versions.* The residual recipe was chosen over 17 versions of the campaign. M1 is a confirmatory test of that recipe on fresh data, and multiplicity is accounted for over the registry, but selection effects cannot be fully excluded. The registry counts in this draft predate the v17x robustness stages, whose rows are still to be added.
8) *Definitions.* Conclusions about multi-process HARQ depend on the PF-averaging definition (Section VII). We use the scheduling-time definition throughout.
9) *Learned baselines.* No learned LA scheme from the literature (for example LOLLA or SALAD) was re-implemented in our simulator.

IX. Conclusion

We asked when deep RL helps downlink link adaptation once it is compared with rules calibrated on the learner's own cues. In one simulator with multi-process HARQ, the answer is: rarely, and by little. On full-buffer goodput the learner adds nothing. With deadline traffic and delayed HARQ feedback a residual A2C policy beats OLLA clearly and beats re-calibrated rules at trace level, but its margin over a carefully calibrated table that combines the same cues is small, depends on the training seed, and is negative for slow users. Most of the apparent advantage over standard baselines came from comparator calibration. We recommend that learned link-adaptation studies report calibrated comparators with an information table, disjoint splits with common random numbers, seed-level intervals and results by mobility regime. A pre-registered replication on a non-oracle base rule will test the most promising configuration found here.

Code, data and pre-registration availability

[Draft placeholder] Release DOI for the simulator, traces, evaluation logs, case registry and the frozen pre-registration texts (with their SHA-256 hashes and freeze commits).

ACKNOWLEDGMENTS

[Draft placeholder] Funding, and the AI-assistance disclosure in the wording required by the target journal (see `submission_checklist.md`).

APPENDIX A
TRAINING DETAILS

Table VII lists the learner's hyper-parameters. They are read from the simulator code by the number-generation script. The same values were used in M1–M4, W1, G6 (which re-uses the M1 policies) and NR1.

TABLE VII
HYPER-PARAMETERS OF THE SHARED MULTI-USER A2C (ALL CASES).

Item	Value
Actor network	57-128-96-3 (residual)
Parameters, MACs per decision	20 099, 19 872
Actor / critic learning rate	3×10^{-4} / 1×10^{-3}
Discount factor γ	0.95
n-step return	10
Entropy weight (start $\rightarrow$ end)	0.03 $\rightarrow$ 0.001
Validation	every 50 episodes on 10 traces
Initial logit bias on $\delta = 0$	3
Deviation penalty	0.02 $\lvert\delta\rvert$
Shaping constant c	0.5
Episodes (deadline / goodput)	3000 / 800
Training seeds per case	10

APPENDIX B
COMPARATOR TABLES FOR M2–M4

Tables VIII–X give the margin of the primary residual A2C scheme over each v17 comparator, in the format of Table V. On-time ratio is the metric in M2, goodput in M3 and M4.

TABLE VIII
M2 (DEADLINE, RTT 0): RESIDUAL A2C VS. EACH COMPARATOR.

Comparator	Margin	90 % CI	Holm p	Beats
Baseline	+4.73	[+4.40, +5.06]	2.3×10^{-63}	yes
OLLA	+2.50	[+2.23, +2.77]	8.6×10^{-39}	yes
OLLA (tuned target)	+2.50	[+2.23, +2.77]	8.6×10^{-39}	yes
Multi-OLLA	+2.30	[+2.05, +2.55]	8.6×10^{-39}	yes
Fixed	+0.93	[+0.72, +1.13]	1.1×10^{-10}	yes
FixedPU (class oracle)	+1.00	[+0.80, +1.20]	9.4×10^{-15}	yes
Context table	+0.43	[+0.21, +0.64]	1.7×10^{-2}	yes
Bandit (global)	+12.57	[+12.09, +13.06]	3.8×10^{-82}	yes
Bandit (delay class)	+12.57	[+12.09, +13.06]	3.8×10^{-82}	yes

APPENDIX C
G6 CALIBRATION DETAILS

Calibration set. 500 traces (seeds 600001–600500) with their own CRN draws, disjoint from the M1 evaluation traces. Every table was written to a file and hashed before the evaluation traces were run.

Single-cue rules. Fixed-500 selects one global offset of -2 CQI steps (the same value as the v17 Fixed rule). Context-500 is the best of all tables over three slack bins (-4, -4, -2). The re-tuned OLLA uses target 0.01 with a 1.0 dB step. FixedPU-500 is a separable search followed by joint refinement.

TABLE IX
M3 (GOODPUT, RTT 8): RESIDUAL A2C VS. EACH COMPARATOR.

Comparator	Margin	90 % CI	Holm p	Beats
Baseline	+8.81	[+7.93, +9.70]	3.8×10^{-44}	yes
OLLA	+1.32	[+0.67, +1.97]	7.2×10^{-3}	yes
OLLA (tuned target)	+2.14	[+1.44, +2.84]	1.9×10^{-5}	yes
Multi-OLLA	+0.20	[$-$0.48, +0.87]	1.0×10^{0}	no
Fixed	$-$0.88	[$-$1.44, $-$0.33]	7.2×10^{-3}	no
FixedPU (class oracle)	0.00	[$-$0.00, 0.00]	1.0×10^{0}	no
Context table	$-$0.88	[$-$1.44, $-$0.33]	7.2×10^{-3}	no
Bandit (global)	+1.32	[+0.67, +1.97]	7.2×10^{-3}	yes
Bandit (delay class)	+1.70	[+1.07, +2.33]	1.0×10^{-4}	yes

TABLE X
M4 (GOODPUT, RTT 0): RESIDUAL A2C VS. EACH COMPARATOR.

Comparator	Margin	90 % CI	Holm p	Beats
Baseline	+7.07	[+6.30, +7.83]	1.5×10^{-38}	yes
OLLA	+0.15	[$-$0.41, +0.72]	1.0×10^{0}	no
OLLA (tuned target)	$-$0.90	[$-$1.50, $-$0.30]	8.2×10^{-2}	no
Multi-OLLA	$-$0.77	[$-$1.35, $-$0.20]	1.3×10^{-1}	no
Fixed	$-$1.73	[$-$2.23, $-$1.24]	1.7×10^{-8}	no
FixedPU (class oracle)	$-$1.04	[$-$1.25, $-$0.83]	4.9×10^{-15}	no
Context table	$-$1.73	[$-$2.23, $-$1.24]	1.7×10^{-8}	no
Bandit (global)	+0.15	[$-$0.41, +0.72]	1.0×10^{0}	no
Bandit (delay class)	+0.51	[$-$0.03, +1.04]	5.9×10^{-1}	no

Combined table. Table XI gives the calibrated Context-joint-500 offsets. Cells with fewer than 200 calibration decisions (marked *) were not searched and keep the composite value of Section V. The search ran from 4 starts (the composite table and three random tables) with at most 6 sweeps each; no start reached the cap.

TABLE XI
CALIBRATED COMBINED TABLE (CONTEXT-JOINT-500): CQI OFFSET PER SLACK BIN b (1: TIGHTEST), DELAY CLASS d, SPEED CLASS v (1: SLOWEST) AND BUSY-HARQ INDICATOR u. *: SPARSE CELL, COMPOSITE VALUE.

		$v=1$		$v=2$		$v=3$	
b	d	$u=0$	$u=1$	$u=0$	$u=1$	$u=0$	$u=1$
1	1	-3^*	-3^*	-4^*	-3	-4^*	-4^*
1	2	-3^*	-3^*	-4^*	-4	-4^*	-4
2	1	-3^*	-4	-4^*	-4	-4^*	-2
2	2	-3^*	-4	-4^*	-3	-4^*	-3
3	1	-1	-1	-2	-1	-2	-1
3	2	-2	-1	-2	-1	-3	-2

Calibration against evaluation. Table XII compares each re-calibrated rule's score on the calibration traces with its on-time ratio on the evaluation traces.

TABLE XII
G6 RULES: CALIBRATION OBJECTIVE AND EVALUATION ON-TIME RATIO.

Rule	Calibration	Evaluation
Combined table (Context-joint-500)	0.6088	0.6086
Context-500	0.5982	0.6040
Fixed-500	0.5953	0.6014
FixedPU-500	0.5914	0.5965
Composite (start of the search)	0.6003	–

APPENDIX D
REGISTRY SUMMARY

Table XIII summarises the case registry by campaign version. A case is one configuration evaluated on its own traces; a row is one scheme evaluated in a case. The full registry is in the supplement. The v17x robustness stages (W1, G6, NR1, G10, A6) are not yet included.

TABLE XIII
CASE REGISTRY BY VERSION: CASES, SCHEME EVALUATIONS (ROWS), CASES WITH A VALID WIN, AND SUPERSEDED CASES.

Version	Cases	Rows	WIN	Superseded
v9	18	18	0	0
v10	15	15	2	0
v11	10	17	0	0
v12	2	2	0	0
v13	3	6	2	0
v14	6	12	3	0
v15	3	6	0	3
v16	6	20	1	4
v17	4	12	2	0
all	67	108	10	7

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